Revision 2026

Introduction to Polymer Science

Polymer Technology · Semester I · Practicum · Credits 4.5 · Contact Hours 90

This is a full handbook-style HTML learning resource built strictly from the supplied curriculum PDF and expanded into a textbook-like format for first-semester diploma students.

Course Code 1091 | **Theory 20 CIE + 60 ESE** | **Practical 20 CIE + 40 ESE** | **Total Marks 140** | **Malayalam Support**

Covered in this handbook

- Course information and outcomes
- Four complete modules
- Practical modules
- Self-learning activities
- Textbooks, references, web resources
- Assessment and evaluation
- Large question bank with answers
- Quick revision and glossary

Course Information

Course Description

Polymers are crucial to modern life because they are lightweight, durable, and versatile. They are used in clothing, packaging, medical devices, vehicles, electronics, and many other products. The course introduces different types of polymers, their chemical structure, properties, and applications.

Students also prepare an album of polymer products available in the market and identify the polymeric material used in each product. This practical orientation makes the course useful for industry awareness and material identification.

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> □ course-□ polymers-□□□□ structure, properties, applications □□□□□□ □□□□□□□□□□□. Market-□□□□□ polymer products □□□□□□□□□□□□□□ album □□□□□□□□□□□□□□ □□□ □□□□□□□□□□□□□□.

Course Objectives

1. To summarise the history, general characteristic and classification of polymers.
2. To understand Natural rubber and different forms of natural rubber.
3. To understand about the commercially available Synthetic rubbers.

4. To understand about different types of plastics and fibres.

Course Outcomes (COs)

CO	Outcome	Level
CO1	Explain history, general characteristic & classification of polymers	Understand
CO2	Explain structure, property & applications of Natural rubber	Understand
CO3	Comprehend synthetic rubbers – structure, properties & applications	Understand
CO4	Explain classification, structure, property & applications of Plastics & fibres	Understand

Prerequisites

Topic	Level
Hydrocarbons, alkane, alkene, alkynes	Secondary Education
Polymers	Secondary school

Teaching Scheme

L	T	P	S	SS/Week	Hours	Credits
3	0	3	0	6	90	4.5

CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	–	–	–	–	–	–	–	–	–	3
CO2	3	–	–	–	–	–	–	–	–	–	3
CO3	3	–	–	–	–	–	–	–	–	–	3
CO4	3	–	–	–	–	–	–	–	–	–	3
Total	12	–	–	–	–	–	–	–	–	–	12

Legends: 1–Low · 2–Medium · 3–High · – No Correlation

Examination Scheme (Total 140)

	CIE	ESE	Total
Theory	20	60	80
Practical	20	40	60
Grand Total			**140**

Module I — History of Polymers and Classification of Polymers

Theory 12 h · Practical 12 h · Mapped CO: CO1

1.1 History of Polymers

Polymers have been used by humans for thousands of years long before their chemical nature was understood. Natural materials such as cotton, wool, silk, wood and natural rubber were the earliest polymer materials.

- **Ancient times** — Mesoamerican civilisations used natural rubber for balls and waterproofing. Cotton, wool and silk were used for textiles.
- **1839** — Charles Goodyear discovered vulcanisation of natural rubber with sulphur, making rubber useful over a wide temperature range.
- **1860s–1900s** — Cellulose nitrate (Parkesine, Celluloid) and regenerated cellulose (Rayon) marked the start of man-made polymers.
- **1907** — Leo Baekeland invented Bakelite, the first fully synthetic thermosetting plastic.
- **1920s–1930s** — Hermann Staudinger proposed the macromolecular hypothesis. Wallace Carothers developed Nylon at DuPont.
- **1930s–1950s** — Commercialisation of polyethylene, PVC, polystyrene and synthetic rubbers (SBR, NBR, CR).
- **1950s onwards** — Stereoregular polymers (Ziegler–Natta), engineering plastics, carbon & aramid fibres, biomedical polymers.

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1.2 Hydrocarbons

Definition: Compounds containing only carbon and hydrogen.

Saturated Hydrocarbons

Only single C–C bonds. General formula of alkanes: C_nH_{2n+2}. Less reactive. Examples: Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8).

Unsaturated Hydrocarbons

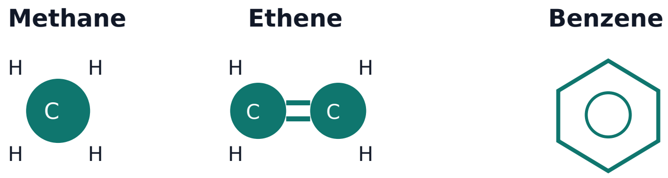
Contain C=C (alkenes) or C≡C (alkynes). More reactive. Examples: Ethene (C_2H_4), Ethyne (C_2H_2).

Aliphatic Hydrocarbons

Open-chain or non-aromatic cyclic compounds (methane, ethene, cyclohexane).

Aromatic Hydrocarbons

Contain benzene rings with delocalised π-electrons. Examples: Benzene, Toluene, Naphthalene.



Structural sketches of methane, ethene and benzene

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1.3 Homologous Series · Alkanes · Alkenes · Alkynes

Homologous series: Organic compounds with the same functional group and similar chemical properties; successive members differ by $–CH_2–$.

Alkanes

C_nH_{2n+2}. Only single bonds. sp^3, bond angle ≈109.5°. Relatively inert. Examples: CH_4, C_2H_6, C_3H_8.

Alkenes

C_nH_{2n}. One C=C. sp^2, planar. Undergo addition. Examples: Ethene, Propene. Many polymer monomers are alkenes.

Alkynes

$C_nH_{2n–2}$. One C≡C. sp, linear. Highly reactive. Example: Ethyne (acetylene).

1.4 Micromolecules & Macromolecules

Micromolecules: Small molecules, MW usually < 1000 g/mol (NaCl 58.5, water, glucose, monomers).

Macromolecules: Very large molecules, MW thousands to millions (NR ≈ 10^5–10^6, proteins, cellulose, plastics). The large size produces strength, elasticity and special physical properties.

1.5 Monomers, Oligomers, Polymers, Repeating Units

- **Monomer** — Small molecule that can join to form a polymer (ethene, vinyl chloride, isoprene, styrene).
- **Oligomer** — Short chain of a few (2–10) monomer units.
- **Polymer** — Large molecule of many repeating units (DP usually > 100).
- **Repeating unit** — The structural unit that repeats; for PE it is $–CH_2–CH_2–$.

$n\ CH_2{=}CH_2 \rightarrow [-CH_2-CH_2-]_n$ (Polyethylene)

1.6 Polymerisation & Degree of Polymerisation

Polymerisation: Process by which monomers join to form a polymer.

- **Addition (chain-growth)** — No loss of atoms (PE, PS, PVC).
- **Condensation (step-growth)** — Small molecule (usually H_2O) eliminated (Nylon, polyester).

Degree of Polymerisation (DP): Average number of repeating units in the chain.

$$M_w \text{ (polymer)} = DP \times M_0 \text{ (repeating unit)} \quad | \quad DP = M_w / M_0$$

Worked Example 1

PE sample M_w = 84 000 g/mol; repeating unit = 28 g/mol. DP = 84 000 / 28 = **3000**.

Worked Example 2

Polystyrene DP = 500; repeating unit = 104 g/mol. M_w = 500 × 104 = **52 000 g/mol**.

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1.7 Classification of Polymers

By Origin

- **Natural** — NR, cellulose, starch, proteins, silk, wool.
- **Synthetic** — PE, PVC, Nylon, SBR, Bakelite.
- **Derived (semi-synthetic)** — Cellulose acetate, vulcanised rubber, rayon.

By Backbone

- **Organic** — Carbon main chain (most commercial polymers).
- **Inorganic** — Non-carbon main chain (Silicone rubber –Si–O–).
- **Homochain** — All atoms in chain same (PE, PS).
- **Heterochain** — Different atoms in chain (Nylon C+N, Polyester C+O, Silicone Si+O).

By Monomer Type

- **Homopolymer** — One monomer (PE, PVC, NR).
- **Copolymer** — Two or more monomers.
  - Random — monomers arranged randomly (SBR).
  - Alternating — regular A–B–A–B sequence.

By Architecture

- **Linear** — Continuous chain, packs well, higher crystallinity (HDPE).
- **Branched** — Side chains, lower density (LDPE).
- **Cross-linked** — 3-D network, does not melt or dissolve (vulcanised rubber, Bakelite).

Linear · Branched · Cross-linked chains

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Origin: Natural / Synthetic / Derived. Structure: Linear / Branched / Cross-linked. Homopolymer = □□□ □□□□□□; Copolymer = □□□□□ □□□□□□.

Practical Module I — Specimen Collection & Album

Objective: Collect natural, synthetic, organic and inorganic polymer samples; record name, 3 properties and applications; prepare an album.

Category	Samples	Typical Properties	Applications
Natural	Cotton, wood, NR band, silk	Biodegradable, renewable	Textiles, paper, tyres
Synthetic	PE bag, PVC pipe, Nylon line, Polyester cloth	Durable, processable	Packaging, pipes, clothing
Organic	PE, PP, Nylon, Polyester, PVC	Carbon backbone	Everyday plastics & fibres
Inorganic	Silicone rubber (spatula, sealant, tubing)	Heat resistant, flexible	Seals, medical, cookware

Album page format

1. Photo / glued sample
2. Polymer name & category
3. Three key properties
4. Two applications
5. Source of sample

Precautions: Collect only clean, dry, non-hazardous samples. Label clearly. Avoid sharp or contaminated pieces.

Module II — Natural Rubber

Theory 11 h · Practical 11 h · Mapped CO: CO2

2.1 Origin, Monomer & Structure

Source: Latex of *Hevea brasiliensis*. Kerala is India’s leading producer. Rubber Board (Kottayam) supports research, quality and marketing.

Monomer: Isoprene (2-methyl-1,3-butadiene), C_5H_8.

Polymer: cis-1,4-Polyisoprene. Nearly 100 % cis configuration gives high elasticity. Molecular weight typically 10^5–10^6 g/mol.

$$n\ CH_2{=}C(CH_3){-}CH{=}CH_2 \rightarrow [-CH_2{-}C(CH_3){=}CH{-}CH_2-]_n \quad \text{(cis-1,4-polyisoprene)}$$

2.2 Properties of Natural Rubber

1. High elasticity and resilience
2. Good tensile strength (after vulcanisation)
3. Excellent abrasion resistance
4. Low hysteresis (good for dynamic uses)
5. Gas permeable; attacked by oils, solvents and ozone (needs protection)

2.3 Latex Forms & Preservation

Field Latex

Fresh from tree. DRC ≈ 30–35 %. Contains proteins, resins, sugars. Unstable; coagulates on standing.

Concentrated Latex

DRC ≈ 60 % by centrifuging or creaming. Easier to transport and process.

Double Centrifuged

Centrifuged twice; higher purity. Used for medical gloves and high-quality goods.

Preservation: Ammonia (0.2–0.7 %) is the main preservative; prevents bacterial action and coagulation. Low-ammonia systems use secondary preservatives (TMTD, ZnO etc.).

Concentration methods: Centrifuging (high-speed separation) and Creaming (creaming agents such as ammonium alginate).

2.4 Forms of Dry Rubber — RSS, TSR, ISNR

- **RSS (Ribbed Smoked Sheet)** — Coagulated, sheeted, ribbed, smoked. Graded visually RSS 1–5.
- **TSR (Technically Specified Rubber)** — Graded by dirt, ash, nitrogen, volatile matter, plasticity. Consistent quality.
- **ISNR (Indian Standard Natural Rubber)** — Indian TSR grades (ISNR 5, 10, 20, 50) under BIS.
- Other forms: Crepe rubber, ADS (Air Dried Sheet).

RSS Production (outline)

1. Dilution & sieving of latex
2. Coagulation with formic/acetic acid
3. Milling into sheets → ribbing
4. Smoking 4–7 days → grading & packing

ISNR / TSR (simplified)

1. Coagulation → crumb
2. Washing & drying → blending to specs → baling

2.5 Cis-NR vs Trans-NR (Gutta-percha)

cis-1,4-Polyisoprene (NR) — Soft, highly elastic at room temperature.

trans-1,4-Polyisoprene (Gutta-percha / balata) — Harder, more crystalline, lower elasticity. Used historically for golf balls, insulation, dental work.

Geometric isomerism around the double bond causes the large property difference.

2.6 Latex Industries & Kerala Rubber Industry

Kerala produces the major share of Indian NR. Important product groups:

- **Latex products** — gloves, balloons, catheters, condoms, foam, adhesives.
- **Dry rubber products** — tyres, tubes, Hawaii slippers, belts, hoses, moulded goods.

Rubber Board (Kottayam) provides technical support and market promotion. Many SMEs in Kottayam, Ernakulam and Pathanamthitta process latex and dry rubber.

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> NR □□□□□□□□□□□ Hevea □□□□□□□ □□□□□□□□□ □□□□□□□□□□ □□□□□□. □□□□□ □□□□□□□□□□ □□□□□□ □□□□□□ □□□□□□□□. Rubber Board □□□□□□□□□□. cis = elastic; trans = gutta-percha. RSS, TSR, ISNR □□□□□□ dry forms.

Practical Module II

- Prepare / observe RSS and ISNR samples; note visual & tactile differences.
- Demonstrate coagulation of NR latex (acid → gel → sheet).
- Demonstrate preservation of NR latex (ammonia, stability observation).
- Album of 6 NR products: Tyre, Inner tube, Latex gloves, Balloon, Hawaii slippers, Latex foam/sponge — record properties & applications.

Lab Series Exam – I covers the above (observation, procedure, viva).

> **Precautions:** Handle ammonia in ventilated area. Use dilute acids; wear gloves & goggles. Dispose latex waste properly.

Module III — Synthetic Rubber

Theory 11 h · Practical 11 h · Mapped CO: CO3

3.1 Introduction

Synthetic rubbers are elastomers made from petrochemical monomers. They overcome NR limitations (supply, oil resistance, temperature range, etc.).

General purpose (GP): SBR, BR, IR — large-volume tyre & general goods.

Special purpose (SP): NBR (oil), CR (weather/flame), IIR (gas barrier), EPDM (weather), CSM, Silicone (extreme temperature).

3.2 IR — Isoprene Rubber

Raw material: Isoprene (C5 stream).

Structure: Predominantly cis-1,4-polyisoprene (NR-like).

5 properties: High resilience, good tensile strength, low hysteresis, good building tack, similar to NR.

Applications: Tyre sidewalls/carcass, medical goods, adhesives, footwear.

3.3 SBR — Styrene-Butadiene Rubber

Raw materials: Styrene + Butadiene (random copolymer, ~23–25 % styrene for tyres).

5 properties: Good abrasion, good aging, moderate resilience, lower cost than NR, processable.

Applications: Passenger tyre treads, belts, footwear, hoses, adhesives.

3.4 BR — Butadiene Rubber

Raw material: 1,3-Butadiene.

Structure: High-cis or medium-cis polybutadiene.

5 properties: Excellent resilience, low T_g, good abrasion, low heat build-up, poor wet grip (usually blended).

Applications: Tyre treads/sidewalls (blends), golf balls, impact modifiers.

3.5 IIR — Butyl Rubber

Raw materials: Isobutylene + small isoprene.

Structure: Mostly saturated polyisobutylene with occasional unsaturation for cure.

5 properties: Outstanding gas impermeability, good weather/ozone resistance, high damping, chemical resistance, low resilience.

Applications: Inner tubes, tubeless liners, pharma stoppers, vibration mounts, sealants.

3.6 EPDM

Raw materials: Ethylene + Propylene + diene (ENB/DCPD).

Structure: Saturated main chain + side-chain unsaturation for vulcanisation.

5 properties: Excellent ozone/weather/heat resistance, good electrical insulation, water resistance, moderate oil resistance, colourable.

Applications: Automotive weather-strips, roofing membranes, cable insulation, radiator hoses, window seals.

3.7 NBR — Nitrile Rubber

Raw materials: Acrylonitrile + Butadiene (ACN 18–50 %).

Structure: Random copolymer; higher ACN → better oil resistance.

5 properties: Excellent oil/fuel resistance, good abrasion, moderate heat resistance, poor ozone resistance, good tensile strength.

Applications: Oil seals, O-rings, fuel hoses, gaskets, printing rolls, industrial gloves.

3.8 CR — Chloroprene Rubber (Neoprene)

Raw material: Chloroprene (2-chloro-1,3-butadiene).

Structure: Predominantly trans-1,4-polychloroprene.

5 properties: Good oil resistance, excellent weather/ozone/flame resistance, good mechanical properties, moderate heat resistance, self-extinguishing tendency.

Applications: Wetsuits, cable jackets, gaskets, adhesives, belts, hoses, vibration mounts.

3.9 CSM — Chlorosulphonated PE

Raw material: PE modified by chlorination + chlorosulphonation.

Structure: PE backbone with Cl and $–SO_2Cl$ groups.

5 properties: Outstanding ozone/weather resistance, good chemical resistance, colour retention, flame resistance, moderate oil resistance.

Applications: Cable sheathing, coated fabrics, pond liners, automotive hoses, protective coatings.

3.10 Silicone Rubber

Raw material: Organosiloxanes (dimethylsiloxane etc.).

Structure: –Si–O– inorganic backbone with organic side groups (usually methyl).

5 properties: Extreme temperature range (–60 °C to +250 °C), excellent weather/ozone resistance, physiological inertness, good electrical insulation, moderate mechanical strength.

Applications: Medical implants & tubing, kitchenware, extreme-environment seals, electrical insulation, baby products, aerospace seals.

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> GP: SBR, BR, IR. SP: NBR (oil), CR (weather+flame), IIR (gas barrier), EPDM (weather), Silicone (extreme temp). □□□□□□□□□□□□□□ □□□□□□□□ property □□□□□□□□□□ application □□□□□□□□□□□□□□□.

Comparison Table — Major Synthetic Rubbers

Rubber	Key Strength	Main Limitation	Typical Use
SBR	Abrasion, cost	Resilience < NR	Tyre treads
BR	Resilience, low T_g	Wet grip	Tyre blends
IIR	Gas barrier	Low resilience	Inner liners
EPDM	Weather/ozone	Oil resistance	Weather-strips
NBR	Oil/fuel	Ozone	Seals, hoses
CR	Weather + oil + flame	Cost	Wetsuits, cables
Silicone	Temperature extremes	Strength, cost	Medical, high-T seals

Practical Module III

Collect product samples of SBR, BR, IIR, EPDM, NBR, CR, CSM and Silicone rubber. For each: label polymer name, list 3 properties and 3 applications. Prepare a neat album and observe differences in feel, flexibility and typical end-use.

Module IV — Plastics and Fibres

Theory 11 h · Practical 11 h · Mapped CO: CO4

4.1 Thermoplastics vs Thermosets

Thermoplastics

Soften on heating, harden on cooling; process is reversible. Linear/branched chains held by secondary forces.

5 examples: PE, PP, PVC, PS, Nylon (also PET, ABS).

Thermosets

Irreversible cross-linking on curing; cannot be remelted. 3-D network.

5 examples: Bakelite, Melamine-formaldehyde, Urea-formaldehyde, Epoxy, Unsaturated polyester.

Property	Thermoplastics	Thermosets
Softening	Reversible	Irreversible (degrade)
Solubility	Often soluble	Insoluble

Recyclability	Generally recyclable	Difficult
High-T dimensional stability	Lower	Higher
Impact resistance	Often tougher	Often more brittle

Advantages of plastics: Lightweight, corrosion resistant, easy moulding, electrical insulation, wide property range, low cost for many grades.

Disadvantages: Most non-biodegradable, limited high-T performance, environmental concerns (waste, microplastics), toxic fumes on burning for some types.

4.2 Household Plastics

Kitchen & Food Storage

- Water/milk bottles → PET, HDPE
- Food containers → PP, HDPE, LDPE
- Bags & wrap → LDPE, PVC cling film
- Cutlery, microwave dishes → PP, PS
- Caps, dustbins, buckets, straws → PP, HDPE, PS

Bathroom & Personal Care

- Toothbrush → PP handle, Nylon bristles
- Cosmetic containers → PET, PP, ABS
- Shower curtains → PVC, PE
- Shampoo bottles → HDPE, PET, PP

Cleaning & Storage

- Spray bottles → PET, HDPE, PP
- Trash bags → LDPE, HDPE
- Bins, hoses, pipes → PP, HDPE, PVC, PE

Electronics, Toys, Furniture

- Phone covers → TPU, PC, ABS, Silicone
- Toys → ABS, PE, PP
- Pens, notebook covers → PP, ABS, PS
- Chairs, tables, crates → PP, HDPE, ABS
- Insulation sheets → EPS, XPS, PU foam

4.3 Fibres

Definition: Long, thin, flexible structures with high length-to-diameter ratio used for textiles, ropes and composites.

Natural: Cotton, wool, silk, coir, jute, arecanut fibre, flax.

Synthetic: Nylon, Polyester, Acrylic, PP, PE (UHMWPE), Carbon fibre, Glass fibre, Aramid.

Fibre	Key Properties	Applications

Wool	Warm, resilient, moisture absorbing	Clothing, carpets, blankets
Silk	Lustrous, strong, smooth	Luxury textiles, sutures
Coir	Coarse, durable, salt-water resistant	Mats, ropes, geo-textiles
Arecanut	Natural, biodegradable	Composite boards, eco-products
Nylon	High strength, abrasion resistant, elastic	Apparel, ropes, tyres, carpets
Acrylic	Wool-like, colourfast, light	Sweaters, blankets, awnings
PE / PP	Light, chemical resistant, low cost	Ropes, non-wovens, geotextiles
Carbon	Very high strength & stiffness, light	Aerospace, sports, auto composites
Glass	High strength, insulating, low cost	Composites, insulation, PCBs

4.4 Liquid Resins, Adhesives & Sealants

- **Liquid resins** — Reactive liquids that cure to solid. Examples: Epoxy, unsaturated polyester, polyurethane resin.
- **Adhesives** — Join surfaces. Examples: PVA (white glue), epoxy adhesive, cyanoacrylate (super glue), rubber-based adhesives.
- **Sealants** — Fill gaps, stop fluid/gas passage. Examples: Silicone sealant, polyurethane sealant, acrylic sealant.

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Thermoplastics = [illegible] [illegible] [illegible]. Thermosets = [illegible] [illegible] [illegible]. Fibres: Natural (wool, silk, coir…) & Synthetic (Nylon, Acrylic, Carbon, Glass…). Resins, adhesives, sealants — joining & sealing-[illegible].

Practical Module IV

- Collect 3 raw thermoplastic + 3 thermoset samples; label polymer name.
- Collect 3 product samples each of LDPE, HDPE, PVC; record name, 3 properties, application.
- Collect 3 natural + 3 synthetic fibre samples; label name, 3 properties, application.
- Prepare systematic album.

Lab Series Exam – II covers the above practical work.

Self-Learning Activities (15 weeks · 90 hours)

Module I — Weeks 1–4

1. **W1** Relate molecular structure to strength, flexibility, transparency. Collect 2 natural polymer samples; learn 3 properties & applications.
2. **W2** Illustrate polymer molecules by linking simple units. Collect 2 synthetic polymer samples; study properties & applications.
3. **W3** Explore Bakelite. Collect 5 organic polymer samples; learn properties & applications.

4. **W4** List 10 monomers and their polymers. Collect inorganic polymer (silicone); learn properties & applications.

Module II — Weeks 5–8

5. **W5** Explore latex-yielding trees. Collect RSS & ISNR; compare properties.
6. **W6** Compare creaming vs centrifuging. Study preservatives for NR latex.
7. **W7** Find centrifuging units in Kerala. Visit/study latex & dry-rubber industry. Learn coagulation chemistry.
8. **W8** Learn chemical name & structure of NR. Differentiate latex product vs dry-rubber product.

Module III — Weeks 9–12

9. **W9** Search synthetic-rubber products. Collect SBR & BR samples; study 3 properties + 3 applications.
10. **W10** Learn CSM & Silicone structures. Collect IR & EPDM samples; study properties & applications.
11. **W11** Learn SBR, IR, BR structures. Collect NBR & CR samples; study properties & applications.
12. **W12** Learn EPDM, NBR, CR structures. Collect CSM & Silicone samples; study properties & applications.

Module IV — Weeks 13–15

13. **W13** Find 10 thermoplastic & thermoset products. Collect 3 raw thermoplastic + 3 thermoset samples; study name, properties, applications.
14. **W14** Learn fibre-forming polymer properties. Collect 3 natural fibre samples; study name, properties, applications.
15. **W15** Learn fibre applications. Collect 3 synthetic fibre samples; study name, properties, applications.

Learning Resources

Text Books

#	Title	Author	Publication
1	Polymer Science	Jayadev Sreedhar, V.R. Gowariker, N.V. Viswanathan	New Age International, 2005
2	Text Book of Polymer Science	Billmeyer F.W.	John Wiley (Asia), 2008
3	Man Made Fibres	R.W. Moncrieff	Wiley, 5th ed., 1970

4	Handbook of NR Production	RRI, Kottayam / Rubber Board	—

References & Web

#	Title / URL	Author / Modules
1	Introductory Polymer Chemistry	G.S. Misra · New Age, 2005
2	Introduction to Polymer Science and Technology	Mustafa Akay · bookboon.com
3	Module I	
4	Module III	
5	Module II	
6	All	

Major Equipment / Software: NIL (as per syllabus). Work is sample collection, coagulation/preservation demos and album preparation.

Assessment Methodology — Practicum

Overall Scheme

Component	Details	Duration	Marks	Converted
Attendance	—	—	—	5
CA1	Avg of self-learning activities	—	10	5
CE	Continuous evaluation	—	50	10
CA2	Practical test (1st half)	3 h	40	10
CA3	Practical test (2nd half)	3 h	40	10
CA4	Series Exam (2 modules)	2 h	50	10
CA5	Series Exam (2 modules)	2 h	50	10
ESE Theory	Written	2.5 h	60	60
ESE Lab	Lab examination (internal)	3 h	40	40

Schedule: CA1 by 4th wk · CE by 7th · CA2 by 11th · CA3 by 14th · Series 1 in 8th wk · Series 2 in 15th wk · ESE end of semester.

CIE Practical Evaluation (50)

#	Criteria	Marks
1	Preparation & Punctuality	10
2	Setup & Procedure	10
3	Observation & Recording	5

4	Analysis & Interpretation	10
5	Viva / Concept	10
6	Workmanship & Teamwork	5

Practical Test (40)

Category	Marks
Write-up / Procedure	10
Setup & Execution	10
Observation & Result	8
Viva Voce	8
Record	4

Series Exam (2 h, 50 marks) & ESE Theory (2.5 h, 60 marks)

Series: Any two modules. Part A: 4×1 · Part B: 6×3 · Part C: 4×7 (with choice) = 50.

ESE: All four modules. Per module: Part A 2×1, Part B 2×3 (from 3), Part C 1×7 (choice) → 15 marks × 4 = 60.

Question Bank (with Answers)

Module I

▾ **1. Define polymer and monomer. Give two examples of each. (2)**

Answer: Monomer = small molecule that can join to form a polymer (ethene, isoprene). Polymer = large molecule of many repeating units (polyethylene, natural rubber).

▾ **2. State the relationship between DP and molecular weight. (2)**

Answer: M_w = DP × M_0 where M_0 is molecular weight of the repeating unit.

▾ **3. Distinguish thermoplastics and thermosets with two examples each. (3)**

Answer: Thermoplastics soften reversibly on heating (PE, PVC). Thermosets form permanent cross-links and cannot be remelted (Bakelite, epoxy).

▾ **4. PE sample M_w = 56 000; repeating unit = 28. Calculate DP. (3)**

Answer: DP = 56 000 / 28 = 2000.

▾ **5. Explain classification by structure (linear, branched, cross-linked). (7)**

Answer: Linear — continuous chain, high packing/crystallinity (HDPE). Branched — side chains, lower density (LDPE). Cross-linked — 3-D network, insoluble, non-melting (vulcanised rubber, Bakelite). Include property consequences and sketches.

Module II

▾ 1. What is the monomer of NR? Write its structure. (2)

Answer: Isoprene (2-methyl-1,3-butadiene), $CH_2=C(CH_3)–CH=CH_2$.

▾ 2. Differentiate field latex and concentrated latex. (3)

Answer: Field latex — fresh, DRC ~30–35 %, unstable. Concentrated — DRC ~60 % by centrifuging/creaming, more stable and transportable.

▾ 3. Outline RSS production. (7)

Answer: Dilution → sieving → acid coagulation → milling/ribbing → smoking 4–7 days → grading (RSS 1–5) → packing. Explain each step briefly.

▾ 4. Why is cis configuration important in NR? (3)

Answer: Cis-1,4 allows flexible chains that coil/uncoil easily → high elasticity. Trans (gutta-percha) is more crystalline and less elastic.

Module III

▾ 1. Name two GP and two SP synthetic rubbers. (2)

Answer: GP — SBR, BR (or IR). SP — NBR, CR (or IIR, EPDM, Silicone).

▾ 2. Why is NBR preferred for oil seals? (3)

Answer: Polar acrylonitrile groups resist swelling in non-polar oils/fuels; oil resistance rises with ACN content.

▾ 3. Compare SBR and NR for tyre applications. (7)

Answer: SBR — better abrasion & aging, lower cost, poorer resilience & tack. NR — higher resilience, better tear & tack. Blends are common.

▾ 4. List five properties and three applications of Silicone rubber. (5)

Answer: Properties — extreme T range, weather/ozone resistance, biocompatibility, electrical insulation, flexibility. Applications — medical tubing, high-T seals, kitchenware.

Module IV

▾ 1. Give five examples each of thermoplastics and thermosets. (3)

Answer: TP — PE, PP, PVC, PS, Nylon. TS — Bakelite, melamine-formaldehyde, epoxy, unsaturated polyester, urea-formaldehyde.

▾ 2. Which plastic is common for water bottles and why? (2)

Answer: PET — clear, tough, good gas barrier, recyclable, food-safe.

▾ 3. Distinguish natural and synthetic fibres with examples. (5)

Answer: Natural — cotton (absorbent), wool (warm), silk (lustrous), coir (durable). Synthetic — Nylon (strong/elastic), Polyester (crease-resistant), Acrylic (wool-like), Carbon (high stiffness).

▾ 4. What are liquid resins, adhesives and sealants? Two examples each. (5)

Answer: Liquid resins — epoxy, unsaturated polyester. Adhesives — PVA, cyanoacrylate. Sealants — silicone sealant, PU sealant.

Viva / Lab Sample Questions

- How do you preserve NR latex in the laboratory?
- What acid is commonly used for latex coagulation?
- How can you distinguish LDPE from HDPE by simple observation?
- Why is silicone preferred for medical applications?
- Name the polymer used in most plastic chairs.
- What is the repeating unit of polyethylene?
- Why are thermosets not recycled by simple melting?

Quick Revision

Module I Summary

- Polymer = macromolecule from monomers via polymerisation.
- DP = M_w / M_0.
- Classify by origin, backbone, monomer type, architecture.
- Hydrocarbons: saturated / unsaturated · aliphatic / aromatic.

Module II Summary

- NR = cis-1,4-polyisoprene from Hevea latex.
- Latex: field → concentrated → double centrifuged.
- Preserve with ammonia. Dry forms: RSS, TSR/ISNR.

- Cis = elastic; Trans = gutta-percha. Kerala + Rubber Board central to Indian NR.

Module III Summary

- GP: SBR, BR, IR — tyres & general goods.
- SP: NBR (oil), CR (weather+oil+flame), IIR (gas), EPDM (weather), Silicone (T extremes).
- Always link special property → application.

Module IV Summary

- TP = remeltable; TS = permanent network.
- Household: PET bottles, PP containers, HDPE pipes, PVC pipes, PS cutlery, ABS toys.
- Fibres: natural vs synthetic. Resins / adhesives / sealants for joining & sealing.

Formula · Common Mistakes · Exam Tips

$$M_w = DP \times M_0 \quad | \quad DP = M_w / M_0$$

- Do not confuse cis-NR with trans (gutta-percha).
- Thermoplastics can be reshaped; thermosets cannot — remember with examples.
- For synthetic rubbers link key property to application (NBR→oil, IIR→air retention...).
- Check MW units in numericals. Album: neat labels + 3 properties + applications.
- Read questions carefully — “list five” vs “explain with structure”.

Glossary

- **Monomer** — Small molecule that can polymerise.
- **Polymer** — Large molecule of repeating units.
- **DP** — Average number of repeating units.
- **Homopolymer** — One monomer type.
- **Copolymer** — Two or more monomer types.
- **Vulcanisation** — Cross-linking of rubber (usually sulphur).
- **Latex** — Aqueous dispersion of polymer particles.
- **Coagulation** — Conversion of latex to solid rubber.

- **Thermoplastic** — Softens reversibly on heating.
- **Thermoset** — Forms permanent network on curing.
- **Elastomer** — Polymer that stretches largely and recovers.
- **Fibre** — High aspect-ratio structure for textiles/composites.
- **Repeating unit** — Structural unit that repeats in the chain.
- **Cross-link** — Covalent bridge between chains.

- **T_g** — Glass transition temperature (glassy below it).